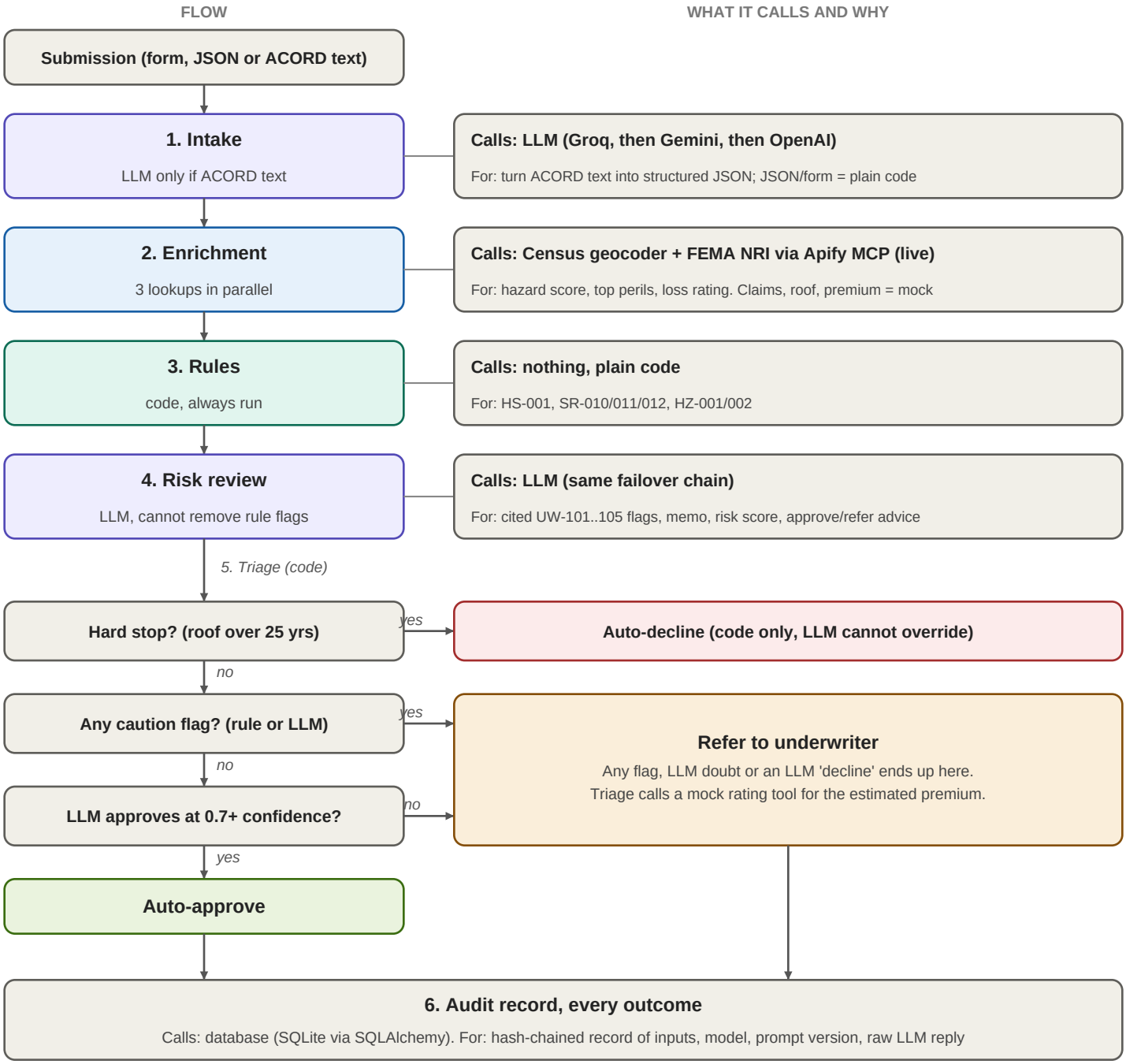
# Underwriter Desk: decision flow

Homeowners underwriting pipeline. Purple steps use an LLM. Everything else is code or data. The LLM can only make a decision more cautious, never less.



## 1. Intake

| Input | What happens | LLM? |
|---|---|---|
| Structured form or JSON | Plain Python maps the fields into a validated Submission. | No |
| **acord_text** (free text) | The LLM converts the text into JSON with a fixed shape (applicant, property, coverage, prior insurance). Each value carries a verbatim quote and a confidence score. Values under 0.6 are dropped, and quotes are checked against the text. The text is treated as data, so instructions inside it are ignored. The result then goes through the same validation as JSON. | Yes |
| Required fields | Property address, Coverage A and effective date. If one is missing the run stops with 'needs more info' and nothing is enriched or decided. | - |

## 2. Enrichment (three lookups run in parallel)

| Source | Type | What it provides | Used for |
|---|---|---|---|
| US Census geocoder | Third party (free, no key) | Lat/lon, matched address, county FIPS | Locating the property. No match raises HZ-002. |
| FEMA National Risk Index via an Apify MCP actor | Third party (live) | a. Composite hazard score (national percentile) and risk tier<br>b. Top perils, each with score and rating (e.g. Ice Storm, Cold Wave)<br>c. Expected annual loss rating, resolution used (tract or county), tract/county FIPS, state, NRI version, geocoded address, Apify run id | HZ-001 (score 70+), HZ-002 (lookup failed), UW-103 (one peril Very High), LLM memo |
| Loss history | **Mock** | Echoes the claims count typed in the submission | SR-011. A real build would call a claims-history vendor. |
| Property data | **Mock** | Roof age computed from the submitted roof year | HS-001 and INFO-001. A real build would call a property data provider. |
| Rating (premium) | **Mock** | Placeholder premium, 0.4% of Coverage A | Shown in the decision. Real premiums come from your rating system. |

## 3. Rules (code, always run)

| Rule | Triggers when | Result |
|---|---|---|
| HS-001 | Roof older than 25 years | Hard stop: auto-decline |
| INFO-001 | Roof 25 years or younger | Info note only |
| SR-010 | Pool present | Caution: refer |
| SR-011 | 2 or more claims in the last 5 years | Caution: refer |
| SR-012 | Trampoline present | Caution: refer |
| HZ-001 | Composite hazard score 70 or higher | Caution: refer |
| HZ-002 | Hazard lookup failed or address not matched | Caution: refer |

Rule flags always stand. The LLM review cannot remove or edit them.

## 4. Risk review (LLM)

**Input:** one JSON with today's date, the submission, all enrichment data and the rule flags already raised.

**What the LLM does**

| Output | Detail |
|---|---|
| Judgment flags | Only the five UW rules below. Each flag must cite a field that exists and has a value, and must not repeat a rule flag. |
| Risk memo | Three to five findings for the underwriter, written from the hazard, claims and coverage data. |
| Risk score | 0 to 100 (higher is riskier). |
| Recommendation | approve, refer or decline, with a confidence (0 to 1) and a short rationale. |

| Rule | The LLM may raise it when |
|---|---|
| UW-101 | Coverage A is under $100,000 or over $5,000,000, or clearly contradicts other submission data. Missing square footage is not a reason. |
| UW-102 | Effective date is in the past or more than 12 months away (must cite the effective date). |
| UW-103 | One peril is rated Very High although the composite score is under the HZ-001 cutoff. |
| UW-104 | Coastal distance is given and is 1 mile or less (must cite that field; never guessed from the address). |
| UW-105 | Submission data is internally inconsistent, e.g. roof older than the house or a year in the future. |

**How it relates to the rules**

• It sees the rule flags in its input and may not repeat, remove or edit them.

• Severity comes from the rule catalog, not from the LLM. Unknown rule ids, rules evaluated by code, and flags citing empty fields are rejected and logged as rejected_flags.

• The prompt asks it to recommend decline on any hard stop and refer on any caution flag or doubt. Code enforces this anyway in triage.

• If the LLM call fails, the run falls back to rules only, with a note in the memo.

## 5. Triage (code makes the decision)

| Order | Check | Outcome |
|---|---|---|
| 1 | Any hard-stop flag (HS-001) | Auto-decline, confidence 1.0. Only code declines; the LLM cannot cause or prevent it. |
| 2 | Any caution flag, from rules or accepted LLM flags | Refer to underwriter. |
| 3 | LLM recommends refer or decline | Refer to underwriter. |
| 4 | LLM approves but confidence is under 0.7 | Refer to underwriter. |
| 5 | No flags and LLM approves at 0.7 or higher | Auto-approve. (With no LLM configured, no flags means approve.) |

## 6. Audit

Every outcome is saved to the database as a hash-chained record: the inputs the agents saw, enrichment data, rule and prompt versions, the model that assessed the risk, the LLM's raw reply and the decision. The chain can be verified and any decision replayed.

## Providers and caveats

• LLM failover order: Groq (gpt-oss-120b), then Gemini (flash), then OpenAI (gpt-5.4-mini). A provider without a key is skipped.

• Loss history, property data and the premium are mocks. Use synthetic applicants only.

• LLM output varies between runs. That is why it can tighten a decision but never loosen one.